# Chapter 10 Notes

## Section 1

### Exercise 1.1

Suppose that $(A, R)$ is isomoprhic to $(A', R')$ under $\phi$. In fact, the biconditional does not need to be proven for any of these - proving the implication also proves the converse.

(a) Suppose $(A, R)$ is proper. Then, for every $x \in A$, $x^*$ is a set. Now consider any $x \in A'$. If $y \in x^*$, then $\phi^{-1}(y) \in \phi^{-1}(x)^*$. And, if $y \in \phi^{-1}(x)^*$, then $\phi(y) \in x^*$. Thus there is a 1-1 correspondence between $x^*$ and $\phi^{-1}(x)^*$, and because $\phi^{-1}(x)^*$ is a set, so is $x^*$ (by the axiom of substitution).

(b) Suppose $(A, R)$ is extensional. Then, for all elements $x, y$ in $A$, if $x \neq y$, then there is at least one element $z$ in $A$ such that $z$ is a component of one of $x$ or $y$, but not of both. Consider any elements $x, y$ in $A'$. If $x \neq y$, then $\phi^{-1}(x) \neq \phi^{-1}(y)$, meaning, because $(A, R)$ is extensional, there is some $z \in A$ such that $z$ is a component of one of $\phi^{-1}(x)$ or $\phi^{-1}(y)$, but not of both. But, because $(A, R)$ and $(A', R')$ are isomorphic, it follows that $\phi(z)$ is a component of $x$ or $y$, but not both, meaning $(A', R')$ is extensional.

(c) Suppose $(A, R)$ is well founded. Thus every subclass of $A$ has an initial element. Consider any subclass $B'$ of $A'$. Then the class $B$ of all $x \in A$ such that $\phi(x) \in B'$ is in a 1-1 correspondence to $A'$. Then $B$ has some initial element $x$. Then $\phi(x) \in B'$. If, for any $y \in \phi(x)^*$, $y \in B'$, we would have $\phi^{-1}(y) \in B$. But $\phi^{-1}(y) \in x^*$, contradicting the fact that $x$ is an intial element of $B$. Hence $\phi(x)^* \cap B = \emptyset$, meaning $(A', R')$ is extensional.

## Section 2

### Notes

Theorem 2.7 requires a bit of clarification. It is remarked that "Since every element of $A$ has $V$-rank then $A$ is well founded (Theorem 3.4 of Chapter 8), hence every element of $A$ has $A$-rank, as we have seen above." This is not fully accurate, as the paragraph above shows that any relational system $(A, \in)$ where $A$ is *transitive* and well founded is such that every element of $x$ has $A$-rank. However, in Theorem 2.7, we are not assuming $A$ is transitive.

To me, it seems that we need to assume that $A$ is transitive.

## Section 3

### Notes

The formulation of the $On$-sequence in the proof of Theorem 3.1 is clearly not correct. Clause (2) reads: "$f_{\alpha+1} = f_\alpha$ (the set of all ordered pairs $\langle x, g(f''_\alpha(x^*))\rangle$ such that $x \in \Gamma_{\alpha+1} - \Gamma_\alpha$)". Obviously, it should instead read: "$f_{\alpha+1} = f_\alpha \cup S$ where $S$ is the set of all ordered pairs $\langle x, g(f''_\alpha(x^*))\rangle$ such that $x \in \Gamma_{\alpha+1} - \Gamma_\alpha$". Here is why this definition is permitted by

Theorem 5.8 (or Theorem 5.7) of Chapter 6. We define $h$ as follows (for all sets). For any $x \in A$, let $R(x)$ be the $\Gamma$ rank of $x$. Also let Func($x$) stand for "$x$ is a function". Then define

$$\forall y(y \in h(x) \equiv (\text{Func}(x) \wedge (y \in x \vee \exists z(y = \langle z, g(h''(x^*))\rangle$$

$$\wedge(\forall z \in \text{Dom}(x)(R(z) < R(x)) \wedge \sim\exists\alpha(\forall z \in \text{Dom}(x)(R(z) < \alpha) \wedge R(x) < \alpha))))))$$

Or, less formally, given any function $x$, $h(x)$ is the set of all $y$ such that either $y \in h(x)$, or $y$ has $\Gamma$ rank $\alpha$, where $\alpha$ is the least ordinal greater than all of the ranks of the elements in the domain of $x$. Then we let $f_{\alpha+1} = h(f_\alpha)$, which clearly works.

It should be noted that Theorem 3.1 also assumes that $\Gamma$ be well founded - this is necessary when using Theorem 2.1 while proving the uniqueness of $F$.

### Exercise 3.1

Recall that Theorem 5.8 of Chapter 6 states: "For any function $h$ defined on all sets, there is a unique function $F$ on $On$ such that for every ordinal $\alpha$, $F(\alpha) = h(F''(\alpha))$." Now, the class of ordinals is well founded, meaning $(On, \in)$ is well founded. Also (as can easliy be seen), every ordinal still has $(On, \in)$ rank $\alpha$. Then, by Theorem 3.1, there is a unique function $F$ on $On$ such that for every $x \in On$, $F(x) = g(F''(x^*)) = g(F''(x))$ (because $x^* = x$ in any relational system $(A, \in)$).

## Section 4

Let us note that if $F$ is a Mostowski-Shepherdson mapping of $A$, and $S \subseteq A$, it does not necessarily follow that $F''(S)$ is a Mostowski-Shepherdson mapping of $S$.

**Lemma 0.1.** For any set $A$ of ordinals, the Mostowski-Shepherdson mapping $F$ on $A$ is such that $F$ maps $A$ onto some ordinal. More specifically, if $A$ is isomorphic to the ordinal $\alpha$, then $F''(A) = \alpha$.

*Proof.* Consider some set $A$ of ordinals. By the counting theorem, $A$ is isomorphic to some ordinal $\alpha$ (let $\phi$ be an isomorphism from $\alpha$ onto $A$). We show that $F''(A) = \alpha$. Because $\phi$ is 1-1 from $\alpha$ onto $A$, $A$ can be put into the sequence $\phi(0), ..., \phi(\beta), ....$ We prove by transfinite induction on all $\phi(\beta) \in A$ that $F(\phi(\beta)) = \beta$, which would mean $F''(A) = \alpha$.

$F(\phi(0)) = 0$ obviously holds.

Suppose that for all ordinals $\beta$ less than $\gamma$ (where $\gamma < \alpha$), $F(\phi(\beta)) = \beta$. Then $F(\phi(\gamma)) = F''(\phi(\gamma) \cap A)$, which is equal to the set of all $F(\phi(\beta))$ where $\beta < \gamma$, and by the assumption, this means $F(\phi(\gamma)) = \gamma$. Thus the proof is complete. □

### Exercise 4.1

We show that for a finite set $A$ of ordinals, the Mostowski-Shepherdson mapping on $A$ maps onto the natural number $n$, where $n$ is the number of elements in $A$. In fact, this is immediate from Lemma 0.1 - we provide an alternative proof below.

*Proof.* Assume $A$ is a finite set of ordinals. It has a Mostowski-Shepherdson map $F$ by Theorem 4.1 (because, for every class $A$ of ordinals, $(A, \in)$ is proper, extensional, and well founded). We show by induction on the size of $A$ that $F''(A) = n$, where $n$ is the number of elements in $A$.

The base case, when $n = 0$, obviously holds.

Suppose the result holds for some $n$. Now consider a set $A$ containing $n + 1$ ordinals. Let $F$ be the Mostowski-Shepherdson mapping of $A$. $A$ must contain some greatest ordinal $\alpha$ (as can easily be seen). Then $A - \{\alpha\}$ has $n$ elements. Also, $F''(A - \{\alpha\})$ is a Mostowski-Shepherdson map of $A - \alpha$. Hence, by the induction assumption, $F''(A - \{\alpha\}) = n$. The only element of $F''(A)$ not in $F''(A - \{\alpha\})$ is $F(\alpha)$. Now, $F(\alpha) = F''(\alpha \cap A) = F''(A - \{\alpha\}) = n$ (because $\alpha$ is the greatest element of $A$). Thus $F''(A) = F''(A - \{\alpha\}) \cup \{F(\alpha)\} = n \cup \{n\} = n^+$, therefore completing the proof. □

## Exercise 4.2

Suppose that $A$ is an infinite set of natural numbers and $F$ is the Mostowski-Shepherdson mapping on $A$. We need to show that $F''(A) = \omega$. We prove this by showing that $F''(A) \subseteq \omega$ and $\omega \subseteq F''(A)$. Like Exercise 4.1, this is in fact immediate from Lemma 0.1. An alternative proof that $\omega \subseteq F''(A)$ is provided below.

*Proof.* On the other hand, we use induction to show $\omega \subseteq F''(A)$.

In the base case, obviously $F(e) = 0$ for the least element $e$ of $A$.

Suppose $m \in F''(A)$ for all $m < n$. So $n - 1 = F(x)$ for some $x \in A$. Because $A$ is infinite, there must be some element $y$ in $A$ such that $y$ is greater than $x$ - let $y$ be the least such element. We show that $F(y) = n$. Well, $F(y) = F''(y \cap A)$.

Clearly $F''(y \cap A)$ contains all natural numbers less than $n$. For, by assumption, all numbers $m$ less than $n$ are $F(z)$ for some $z \in A$. Because $F$ is 1-1, it must be that either $z = x$ or $z \in x$ (because either $F(z) = n - 1$ or $F(z) \in F(x)$); because $x < y$, $z < y$ ($z \in y$, meaning $F(z) \in F''(y \cap A)$).

Next, we need to show that, if $z \in F''(y \cap A)$, then $z$ is a natural number less than $n$. But suppose we had some $z \in F''(y \cap A)$ such that $F(z) \notin \omega$. Now, $z$ must be less than $y$. Hence either $z = x$ or $z < x$. $z \neq x$ because $F(x)$ is a natural number. If $z < x$, then $z \in x$, which would mean $F(z) \in F''(x \cap A) = F(x)$, contrary to the fact that $F(x)$ is a natural number. Thus all members of $F''(y \cap A)$ are natural numbers.

Hence the members of $F''(y \cap A)$ are those and only those natural numbers less than $n$, meaning $F(y) = F''(y \cap A) = n$. Thus the induction is complete.

Hence $\omega = F''(A)$.

□

## Exercise 4.3

Suppose that $A$ is a class of ordinals and $A$ is not a set. Let $F$ be the Mostowski-Shepherdson mapping on $A$. We are to show that $F$ maps $A$ onto the class $On$ of all ordinals. We show this by showing that $F''(A) \subseteq On$ and $On \subseteq F''(A)$.

On the one hand, suppose that $x \in F''(A)$. Then $x = F(y)$ for some $y \in A$. Now, $F(y) = F''(y \cap A)$. Clearly $y \cap A$ is a set of ordinals. Hence, by Lemma 0.1, $F''(y \cap A) = \beta$ for some ordinal $\beta$. Thus $x \in On$, meaning $F''(A) \subseteq On$.

On the other hand, we prove by transfinite induction that $On \subseteq F''(A)$. Note that, because $A$ is a proper class of ordinals, it can be placed into a 1-1 order preserving correspondence with $On$ (under some function $\phi$ from $On$ onto $A$). We show that for every ordinal $\alpha$, $F(\phi(\alpha)) = \alpha$ (thus meaning $On \subseteq F''(A)$). We prove this by transfinite induction.

Obviously $F(\phi(0)) = 0$.

Suppose, for all ordinals $\beta < \alpha$, $F(\phi(\beta)) = \beta$. Then $F(\phi(\alpha)) = F''(\phi(\alpha) \cap A) = F''(S)$, where $S$ is the set of all $F(\phi(\beta))$ for all $\beta < \alpha$. But, by assumption, this contains all ordinals less than $\alpha$. Hence $F(\phi(\alpha)) = F''(S) = \alpha$.

Thus, for every ordinal $\alpha$, $F(\phi(\alpha)) = \alpha$, which clearly means $On \subseteq F''(A)$.

Thus $F''(A) = On$.

# Section 6

## Notes

Note that theorem 6.1, as stated, is not what is proven. Clearly $F$ is assumed to be an $\in$-isomoprhism, though this is not stated.

Note that, in proof 1 of Theorem 6.2, $F''(A)$ is transitive by Theorem 4.1.

In Theorem 6.3, note that $x$ must have some rank by Theorem 2.5. Also, the mention of $P_2$ is actually a reference to $P_8$.